

AIGR Gene Hypothesis Deep Research —

Paramecium tetraurelia A0BFB4

Focus type: function_assignment **Hypothesis slug:** autophagy-ancestry-versus-hmm-family-placement **Target:** A0BFB4 (GSPATT00028266001), 381 aa, *Paramecium tetraurelia* (NCBITaxon:5888) **Source file:** `genes/PARTE/A0BFB4/A0BFB4-ai-review.yaml`

Summary

The seed hypothesis asks whether the four autophagy IBA terms propagated into A0BFB4 through PANTHER family PTHR24348 reflect genuine ATG1/ULK ancestry or a misplacement created by the competing live UniProt/InterPro HMM assignment to the generic family PTHR44167:SF18. **The verdict is partially supported, with a clean split: the ancestry is real and the molecular-function terms should be retained, but the propagated autophagy process/location terms are over-annotations that should be removed or flagged uncertain.**

The apparent family-placement conflict dissolves on inspection. PTHR24348 is officially named "Serine/threonine-protein kinase UNC-51-RELATED" — the ATG1/ULK autophagy-initiating kinase family (UNC-51 is the *C. elegans* ATG1 ortholog). An independent Smith-Waterman local alignment of A0BFB4's kinase domain ranks human ULK2 (35.3% identity) and ULK1 (34.5%) as its closest characterized references, above the STE20-family kinases LOK/SLK and above CHEK2. The "competing" best-hit family PTHR44167:SF18 is a functionally uncharacterized "Serine/Threonine Protein Kinase" catch-all with no assigned GO. So the two families are not truly competing: one captures the evolutionary clade, the other is a scoring artifact carrying no function. A0BFB4 genuinely belongs to the ATG1/ULK clade by ancestry, and this ancestry is **not** a misplacement.

Ancestry, however, does not license the pathway-level terms. A0BFB4 is a bare 381-aa kinase catalytic domain. A comparative-genomics study that explicitly includes *Paramecium tetraurelia* ([PMID: 28123910](#)) shows ciliates do not encode a typical Atg1: every candidate kinase lacks the Atg1-specific C-terminal domain required to build the Atg1 initiation complex, and the complex partners (Atg13, Atg17/FIP200, Atg101) are absent from ciliate genomes. An

AlphaFold structural check (AF-A0BFB4-F1) confirms the missing module is genuinely absent — the only confidently folded non-kinase element is N-terminal, while the C-terminus is a short, partly disordered 33-residue tail. Because the machinery that nucleates a phagophore is absent in the ciliate lineage and in this protein specifically, GO:0000045, GO:0000407, GO:0005776, and GO:0010506 rest solely on phylogenetic IBA inference and cannot be executed by the canonical mechanism. The molecular-function terms (protein Ser/Thr kinase activity, ATP binding) are strongly supported and should be kept.

Executive Judgment

Verdict: Partially supported. Family placement is correct (ATG1/ULK ancestry confirmed by two independent lines); MF terms supported; the four IBA-propagated autophagy BP/CC terms are over-annotations (weakly supported) for this ciliate protein.

Reasoning:

1. **No family misplacement.** The PAINT/TreeGraft graft is into PTHR24348 = "Ser/Thr kinase UNC-51-RELATED" = the ATG1/ULK family. Top sequence identity is to ULK1/ULK2 (~35%), members of that same family. The graft reflects genuine ancestry.
2. **The "competing" HMM is not competing.** PTHR44167:SF18 is a generic, functionally uncharacterized kinase family with no specific GO. The PTHR24348-vs-PTHR44167 discrepancy is a best-hit-vs-graft methodology artifact, not evidence of an alternative function.
3. **Target-relevant biology weakens the specific autophagy function.** Ciliates (including *Paramecium tetraurelia*) lack a typical Atg1: kinase candidates lack the Atg1 C-terminal domain, and the Atg1-complex partners are absent. A0BFB4 fits this class exactly and, per AlphaFold, has no C-terminal partner-binding module. The process/location terms cannot be executed by the canonical mechanism.

Most important caveats: (a) the Atg1-absence evidence is in-silico and does not fully exclude a divergent/non-canonical role; (b) ~35% kinase-domain identity places the protein in the ATG1/ULK clade but is not proof of one-to-one orthology; (c) there are no experimental data on A0BFB4 itself (localization, substrates, knockdown phenotype).

Key Findings

Finding 1 — The autophagy IBA rests on genuine ATG1/UNC-51 ancestry, not a misplacement

The four autophagy IBA terms enter A0BFB4 via a PAINT/TreeGrafter graft into PANTHER family **PTHR24348**, officially "Serine/threonine-protein kinase UNC-51-RELATED." UNC-51 is the *C. elegans* ATG1 ortholog; this family is the ATG1/ULK clade of autophagy-initiating kinases. The seed hypothesis feared that the live UniProt/InterPro HMM best-hit — **PTHR44167:SF18** (match 70–348, reported score 7.6e-61) — represented a competing, contradictory placement. It does not: PTHR44167 carries the current InterPro name "Serine/Threonine Protein Kinase," a functionally uncharacterized catch-all family with **no specific GO or function assigned**. A best-single-HMM assignment to a generic family is a scoring artifact, not evidence of a distinct biological function.

An independent sequence-identity check corroborates the ATG1/ULK reading. A Smith-Waterman local alignment (BLOSUM62, gap -11/-1) of A0BFB4's kinase domain against a panel of characterized human kinases ranked:

Human reference	Family	% identity	SW score
ULK2	ATG1/ULK (PTHR24348)	35.3%	306
ULK1	ATG1/ULK (PTHR24348)	34.5%	299
LOK/STK10	STE20	32.3%	—
SLK	STE20	30.5%	—
CHEK2	CAMK/CHK	25.3%	—
yeast Atg1	ATG1	~23%	—

Two independent lines — the PANTHER tree graft and top-hit sequence identity — agree on ATG1/ULK ancestry. **The PTHR44167 assignment is a generic best-HMM artifact and does not refute the autophagy-family placement.** This reconciles the discrepancy in the seed hypothesis in favor of "confirmed ancestry, not misplacement."

Finding 2 — A0BFB4 is an intact, catalytically active single-domain Ser/Thr kinase (MF terms strongly supported)

A0BFB4 (381 aa) carries only protein-kinase signatures: Pfam PF00069, InterPro IPR000719 (protein kinase domain), IPR017441 (ATP-binding site), IPR008271 (Ser/Thr active site); SMART S_TKc; PROSITE PS00107/PS00108/PS50011. All canonical catalytic residues are present and intact:

- Glycine-rich P-loop: **GEGTFG**
- VAIK β 3 lysine: **CKV**
- HRD catalytic loop: **HRDIK**
- DFG Mg^{2+} -binding motif: present
- APE motif: present

This architecture predicts a catalytically active (non-pseudo) kinase and directly supports **GO:0004674 (protein serine/threonine kinase activity)** and **GO:0005524 (ATP binding)**. Critically, the protein is a *bare catalytic domain*: it lacks the ULK/ATG1 C-terminal (CTD/MIT) module that binds ATG13/FIP200/ATG101 and targets the kinase to the phagophore. The molecular-function annotations are robust; the CC and BP autophagy terms depend on machinery the protein does not carry.

Finding 3 — Comparative genomics: *Paramecium* and all ciliates lack a typical Atg1

Aslan, Küçükoğlu & Arslanyolu (2017, [PMID: 28123910](#); open access PMC5244887) surveyed ATG proteins across five ciliate genomes, **explicitly including *Paramecium tetraurelia*** (plus *Tetrahymena thermophila*, *Ichthyophthirius multifiliis*, *Oxytricha trifallax*, *Stylonychia lemnae*). Two abstract statements directly govern this curation:

"ciliate genomes do not encode typical Atg1 since all the candidate sequences lack an Atg1-specific C-terminal domain which is essential for Atg1 complex formation"

"Consistent with the absence of Atg1, ciliates also lack other members of the Atg1 complex"

The full text adds that "these kinase domain containing proteins should not be considered as Atg1 candidates without further experimental analyses, since they all lack the important C-terminal domains," and that "Other components of this kinase complex (Atg1, Atg13, Atg17,

Atg29 and Atg31) are also absent in ciliates." Canonical Atg1/ULK1 has three regions (N-terminal kinase domain, central Pro/Ser-rich region, C-terminal domain); the C-terminal domain binds Atg13/Atg17(FIP200)/Atg101 and is essential for the complex that induces autophagy. **A0BFB4 is exactly the kinase-domain-only class the authors say should NOT be counted as Atg1 without experiments.** The authors further propose that ciliate autophagosome formation may proceed via Atg6 (Beclin/Vps30) in an *Atg1-independent* manner — meaning even where autophagy occurs in ciliates, this kinase need not be the initiator.

Finding 4 — AlphaFold confirms no C-terminal ATG1 partner-binding domain

The AlphaFold DB model **AF-A0BFB4-F1** (v6) provides a residue-level structural test. Segmenting by the HMM kinase match (residues 70–348):

Region	Residues	Mean pLDDT	% > 70	% < 50	Interpretation
N-terminal flank	1–69	93.2	99%	0%	Confidently folded
Kinase domain	70–348	91.6	95%	—	Confidently folded
C-terminal flank	349–381	64.2	39%	30%	Short, partly disordered
Whole protein	1–381	89.5	—	—	High-confidence model

The only confidently structured non-kinase element is **N-terminal**, whereas the ATG1/ULK1 partner-binding module (recruiting ATG13/ATG17-FIP200/ATG101) is a **C-terminal** domain. A0BFB4 has no such C-terminal domain — its C-terminus is a 33-residue low-confidence tail — and the N-terminal folded region matches no InterPro signature, so its function is unresolved. This independently clinches the over-annotation call: the protein structurally lacks the module that would assemble the Atg1 initiation complex.

Mechanistic Model / Interpretation

ANCESTRY (real)	EXECUTABLE FUNCTION (not supported)
A0BFB4 kinase domain ~35% id to ULK1/ULK2 - grafts into PTHR24348 ▼ (UNC-51/ATG1 family) ATG1/ULK-family Ser/Thr kinase - intact catalytic motifs ▼ (P-loop, VAIK, HRD, DFG, APE) ACTIVE kinase ✓ MF terms - bare catalytic domain ▼ No C-terminal partner-binding module (AlphaFold: C-term = 33-aa disordered tail)	Autophagy initiation requires: • Atg1 C-terminal domain — ABSENT • Atg13 partner — ABSENT in ciliates • Atg17/FIP200 — ABSENT in ciliates • Atg101 — ABSENT in ciliates ↓ No phagophore-nucleation complex ✗ BP/CC autophagy terms cannot execute

How the discrepancy is reconciled. The seed hypothesis framed PTHR24348 (autophagy) vs PTHR44167:SF18 (generic kinase) as competing placements. They are not truly competing: PTHR24348 captures the *evolutionary clade* (ATG1/ULK), and PTHR44167 is a *scoring catch-all* with no function. So the family-placement conflict dissolves — A0BFB4 really is an ATG1/ULK-family kinase by ancestry and by top sequence identity.

Why ancestry does not license the process/location terms. Membership in the ATG1/ULK clade by kinase-domain homology is necessary but not sufficient to perform autophagosome initiation. That function is delivered by the C-terminal scaffolding domain and its binding partners, all of which are absent in ciliates generally and in A0BFB4 specifically (by architecture and by AlphaFold structure). IBA propagation transfers terms based on the phylogenetic node, but the transfer overreaches because it ignores the loss of the accessory module in the ciliate lineage. This is a textbook case where **evolutionary placement is correct but functional propagation is too strong**.

The seed guidance cautions that "family labels, missing animal accessory domains, absent target experiments and short donor lists alone are not evidence of loss." The over-annotation call here does not rest on any single such heuristic. It rests on convergent, target-relevant evidence: (1) a dedicated ciliate-wide comparative-genomics study that names *Paramecium tetraurelia* and states the C-terminal domain and entire Atg1 complex are absent; (2) an AlphaFold structure showing the missing module is genuinely absent, not merely HMM-

undetected; and (3) the mechanistic requirement that the Atg1 complex — not a lone kinase domain — nucleates the phagophore. Together these exceed a "missing accessory domain" argument.

Evidence Base / Evidence Matrix

#	Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confidence & limitations
1	UniProt A0BFB4; InterPro (live)	Computational / database	Qualifies	Live HMM family identity	Best-hit family PTHR44167:SF18 ("Serine/Threonine Protein Kinase", generic, no functional GO)	<i>P. tetraurelia</i>	High confidence for the family; carries functional role
2	PANTHER PTHR24348 (InterPro name)	Structural/ evolutionary	Supports ancestry	Is the PAINT graft into an autophagy family?	PTHR24348 = "Ser/Thr kinase UNC-51-RELATED" = ATG1/ULK family	Cross-species tree	High; placed per PANTHER tree
3	This work (Smith-Waterman, BLOSUM62)	Computational	Supports ancestry	Closest characterized kinase to A0BFB4	ULK2 35.3%, ULK1 34.5% > LOK 32.3% > SLK 30.5% > CHEK2 25.3%	381-aa kinase domain	Model; best-hit over kinase domain; not fully phylogenetic
4	This work (motif scan)	Computational (sequence)	Supports MF	Is it a catalytically active kinase?	Intact P-loop (GEGTFG), VAIK-K, HRD-D, DFG, APE	Sequence	High confidence for "active" kinase; supports MF term
5	PMID: 28123910 (Aslan 2017)	Comparative genomics (primary, in-silico)	Refutes/ qualifies BP/CC	Do ciliates (incl. <i>Paramecium</i>) have a functional Atg1?	No typical Atg1; candidates lack Atg1 C-terminal domain; Atg1-complex members absent; <i>P. tetraurelia</i> is	5 ciliate genomes	Model; high; in-silico; doesn't exclude non-canonical role

#	Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confidence & limitations
					one of 5 genomes		
6	PMID: 24280724 (Akematsu 2014)	Mutant/ pathway (Tetrahymena)	Qualifies (context)	Is ciliate autophagy functional/ Atg1-dependent?	Programmed nuclear death is "a novel type of autophagy"; VPS34/ATG8-driven; "absence of some ATG genes"	<i>T. thermophila</i>	Autophagy exists via no ATG1
7	AlphaFold AF-A0BFB4-F1 (v6); this work	Structural/ computational	Refutes/ qualifies CC/BP	Does A0BFB4 have a C-terminal ATG1 partner-binding domain?	Confident N-term (1-69, pLDDT 93) + kinase (70-348, pLDDT 92); C-term (349-381) short, disordered (pLDDT 64); no C-terminal CTD/ MIT	<i>P. tetraurelia</i> model	High fidelity topology N-term element function unresolved
8	PMID: 21624459	Review	Qualifies (context)	Does autophagy occur in ciliates?	Programmed nuclear death by autophagy is a real ciliate process	<i>Tetrahymena</i>	Oriented only

GO Curation Implications (leads — require curator verification)

GO term	Aspect	Current basis	Lead recommendation	Rationale
GO:0004674 protein serine/threonine kinase activity	MF	IBA	RETAIN	Intact catalytic motifs; strongly supported
GO:0005524 ATP binding	MF	IEA/ architecture	RETAIN	Canonical P-loop/ VAIK ATP site intact
GO:0000045 autophagosome assembly	BP	IBA (PTHR24348)	REMOVE or mark uncertain	Requires Atg1 complex absent in ciliates; no C-terminal domain
GO:0000407 phagophore assembly site	CC	IBA	REMOVE or mark uncertain	Depends on complex recruitment; localization unconfirmed
GO:0005776 autophagosome	CC	IBA	REMOVE or mark uncertain	No structural/ experimental support in <i>Paramecium</i>
GO:0010506 regulation of autophagy	BP	IBA	GENERALIZE / non-core / uncertain	Non-canonical regulatory role conceivable but unproven

Overall, the MF annotations should be retained — and are more informative than a bare "protein binding" placeholder — while the four IBA-propagated autophagy BP/CC terms should be treated as over-annotations pending experimental support, and are candidates for removal or an uncertainty qualifier. If the framework prefers not to delete IBA terms, the minimum defensible action is a caveat noting lineage-specific loss of the accessory module, citing

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and the AlphaFold structure. These are leads, not final GO actions.

Mechanistic Scope

- **Directly testable / supported:** A0BFB4 is an intact, likely catalytically active protein serine/threonine kinase (ATP binding + phosphotransfer). This is the immediate molecular activity and is well supported by domain architecture.
- **Downstream / inferred only:** "autophagosome assembly," "phagophore assembly site," "autophagosome," and "regulation of autophagy" are pathway-level roles that, in yeast/mammals, depend on ATG1/ULK1 assembling with Atg13–Atg17(FIP200)–Atg101 via its C-terminal domain. A0BFB4 lacks that domain and ciliates lack those partners; hence these are inferred pathway consequences of the ancestral family, **not established activities of this gene product**.
- **Localization** (phagophore/autophagosome) is a consequence of targeting via the missing module — not supported.

No *Paramecium*-specific experiment (substrate, localization, mutant phenotype) exists for A0BFB4, so all process/location claims are inference only.

Conflicts and Alternatives

- **Database carry-over / method artifact.** The autophagy terms are IBA propagations from the ATG1/ULK ancestral node; the live HMM lands on a generic family. Both are automated outputs; neither is experimental.
- **Paralog / expansion context.** *Paramecium tetraurelia* underwent successive whole-genome duplications, producing large kinase paralog sets; A0BFB4 may be one of several divergent ATG1-like kinase-domain paralogs, further weakening one-to-one ATG1 orthology.
- **Alternative pathway.** Aslan et al. propose ciliate autophagosome formation may proceed via Atg6/Beclin (Vps30) in an Atg1-independent manner — consistent with *Tetrahymena* programmed nuclear death being VPS34/ATG8-driven ([PMID: 24280724](#)) rather than ATG1-driven.
- **Not a competing function.** The generic PTHR44167 framing is not supported by the current family name; there is no positive evidence for a distinct alternative function (e.g., DNA-damage checkpoint) to replace autophagy.

- **Genuine tension.** Ciliates DO have autophagy machinery (programmed nuclear death;

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— the pathway is not absent wholesale. But the specific Atg1-complex-mediated initiation step that these GO terms describe is the part shown to be absent.

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
No experimental data on A0BFB4	UniProt/literature — none found	All functional calls are inferential	Localization (GFP), in-vitro kinase assay, RNAi autophagy phenotyping in <i>Paramecium</i>
Is A0BFB4 the <i>Paramecium</i> Atg1 candidate in Aslan et al.?	Paper analyzes <i>Paramecium</i> ; individual accessions not extractable here	Exact mapping to A0BFB4	Cross-reference the paper's supplementary accession list
Non-kinase-domain partner-binding capacity	AlphaFold: folded N-term (1-69), short disordered C-term	Determines whether a cryptic module exists	<i>Partly resolved:</i> no C-terminal domain; N-term matches no InterPro — HHpred/Foldseek could assign it
Orthology vs homology	~35% identity; PTHR24348 graft	Determines if function transfer is valid at all	Reciprocal best-hit + gene-tree/species-tree reconciliation
Substrate identity	Kinase active, substrate unknown	Function could be non-autophagy	Phosphoproteomics after knockdown
Full phylogeny vs best-hit	Only pairwise identities (all-pairs tree run timed out)	Firms up clade assignment	ML/NJ tree with ciliate homologs incl. C-terminal-domain character

Discriminating Tests

1. **RNAi/knockdown of A0BFB4 in *Paramecium* + autophagy readout** (ATG8 lipidation/puncta; starvation- or autogamy-induced macronuclear autophagy) — directly tests any autophagy role.
 2. **In-vitro kinase assay** on recombinant A0BFB4 to confirm catalytic activity (validates MF) and, with candidate substrates, probe the pathway.
 3. **Co-IP / proximity labeling** for Atg13/FIP200/Atg101 homologs — tests whether an Atg1 complex can form (predicted negative given their absence).
 4. **GFP localization** under starvation/conjugation — tests "phagophore assembly site"/"autophagosome" CC terms.
 5. **Phosphoproteomics** comparing wild-type vs knockdown to identify substrates and infer the real pathway.
 6. **Phylogenetic placement / Foldseek-DALI** with a curated ATG1/ULK vs CAMK-group alignment across ciliates and model eukaryotes, scoring the C-terminal-domain presence/absence character; structural search of the unassigned N-terminal element.
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Curation Leads (require curator verification)

- **Candidate reference to add:** [PMID: 28123910](#) (Aslan, Küçükoğlu & Arslanyolu, *PeerJ* 2017). Verify snippets: "*ciliate genomes do not encode typical Atg1 since all the candidate sequences lack an Atg1-specific C-terminal domain which is essential for Atg1 complex formation*" and "*Consistent with the absence of Atg1, ciliates also lack other members of the Atg1 complex.*" Full text (PMC5244887) lists *Paramecium tetraurelia* among five genomes and states these kinase-domain proteins "should not be considered as Atg1 candidates without further experimental analyses."
- **Supporting reference:** [PMID: 24280724](#) — verify: "*Programmed nuclear death (PND) in the ciliate protozoan Tetrahymena thermophila is a novel type of autophagy*" and "*the absence of some ATG genes in this organism.*"
- **Structural provenance:** AlphaFold AF-A0BFB4-F1 (v6) — C-terminal region (349–381) mean pLDDT 64.2, no C-terminal partner-binding domain; document as evidence the Atg1 accessory module is genuinely absent.

- **Candidate action changes:** Retain MF (GO:0004674, GO:0005524). Flag/remove or add uncertainty to BP GO:0000045, GO:0010506 and CC GO:0000407, GO:0005776 as ATG1-complex-dependent functions unsupported in ciliates.
- **Suggested curator questions:** (i) Should IBA autophagy terms be down-weighted when the target lineage demonstrably lacks the required complex? (ii) Is a generic "protein phosphorylation" BP a safer core than "autophagosome assembly"? (iii) Should the MF terms be re-sourced to sequence/architecture evidence (ISS/ISM) rather than remain tied to the same IBA graft?
- **Suggested experiments:** the six Discriminating Tests above.

Proposed Follow-up Experiments / Actions

1. **For the curator (immediate):** apply the GO decision table — retain the two MF terms, remove or flag the four autophagy BP/CC terms — and attach

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plus the AlphaFold provenance as justification.

2. **For wet-lab resolution:** RNAi knockdown + autophagy-flux assay under starvation/autogamy is the single most decisive experiment; a negative result would refute the BP terms outright.
 3. **For bioinformatic firming-up:** run an ML phylogeny of ciliate ATG1-like kinases scoring C-terminal-domain presence, and a Foldseek/HHpred search of the unassigned N-terminal folded element to determine whether it confers any lineage-specific targeting or regulatory role.
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Provenance / Artifacts

- Catalytic-motif scan, family reconciliation table, and Smith-Waterman identity table (executed code + outputs).
- AlphaFold AF-A0BFB4-F1 (v6) per-residue pLDDT-by-region computation.
- UniProt/InterPro/PANTHER record retrieval; EuropePMC abstract + full-text extraction confirming *Paramecium* inclusion and Atg1-absence statements.

*Limitations: all computational results are conservative inferences; no experimental data on A0BFB4 exist; the Atg1-absence evidence is in-silico. Local repository *-bioinformatics analyses were withheld from this run.*

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